

Lab 1 – Ideal Gas Law

CHE 260 – Thermodynamics and Heat Transfer

Group Members

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1 Introduction

2 Results and Discussion

2.1 Part 1: Path Independence and Tank Volume Ratio

Part 1 of the lab discusses the comparison between the movement of air between two identical volume tanks, one with a gauge pressure of 40 PSI to another with a gauge pressure of 0-PSI. The main point of comparison is the differences in a slow equalization of pressure versus a rapid equalization of pressure.

2.1.1 Processed Results: Rapid and Slow Expansion

Figures 1–6 show the measured pressure, temperature, and mass-flow histories for the rapid and slow expansions.

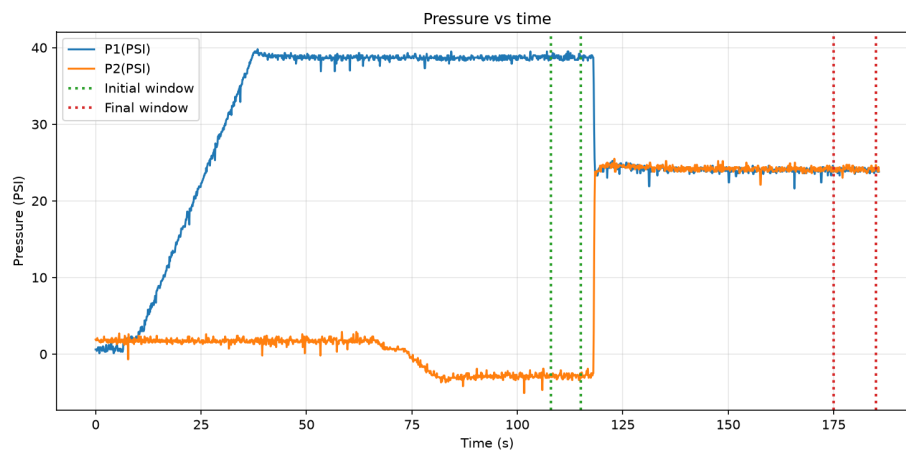


Figure 1: Tank pressures during the Part 1a rapid expansion.

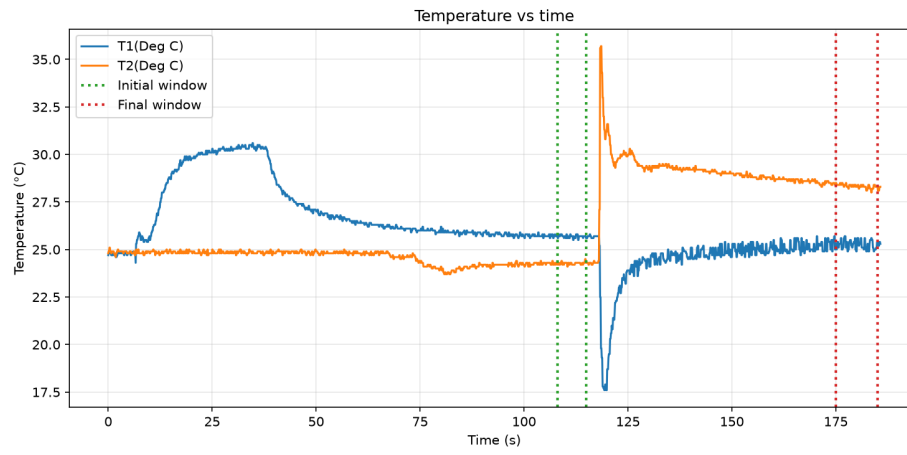


Figure 2: Tank temperatures during the Part 1a rapid expansion.

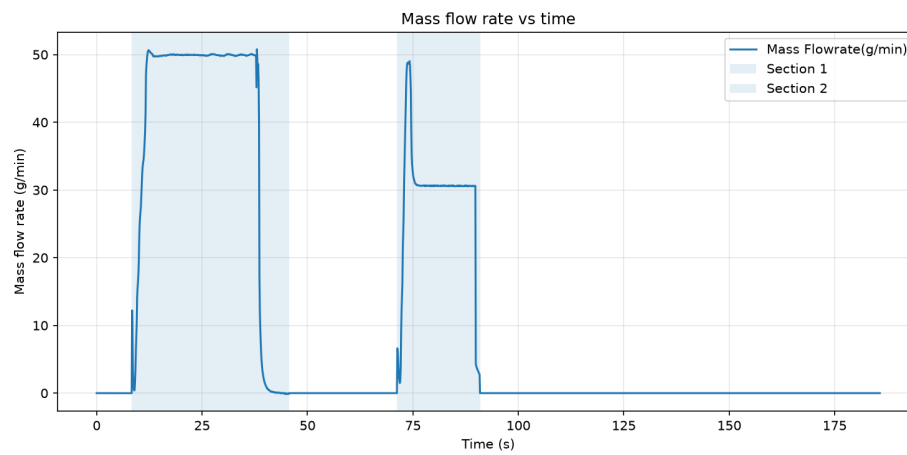


Figure 3: Mass flow rate during the Part 1a rapid expansion.

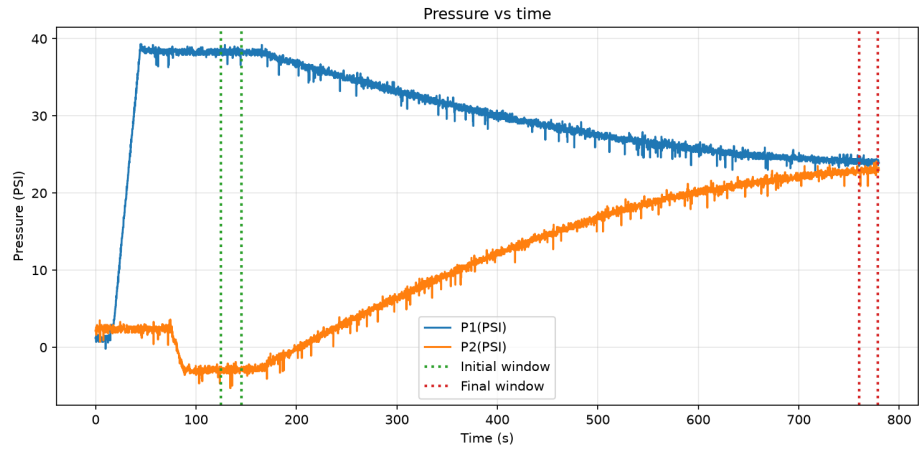


Figure 4: Tank pressures during the Part 1b slow expansion.

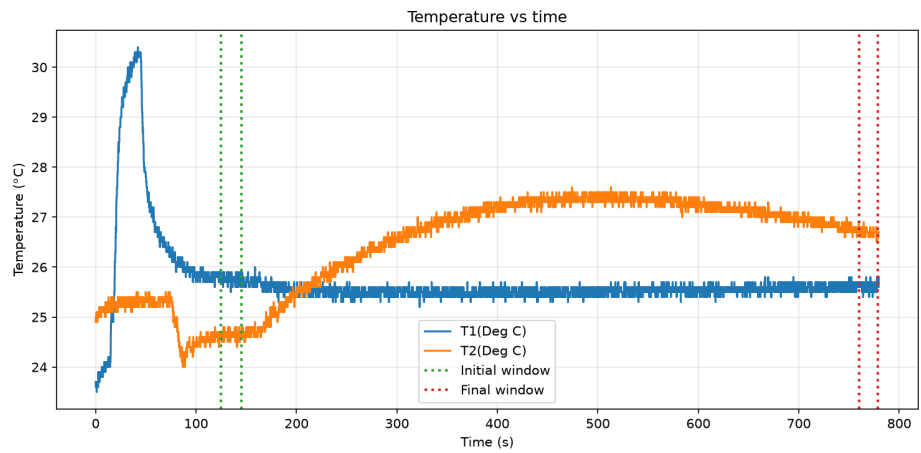


Figure 5: Tank temperatures during the Part 1b slow expansion.

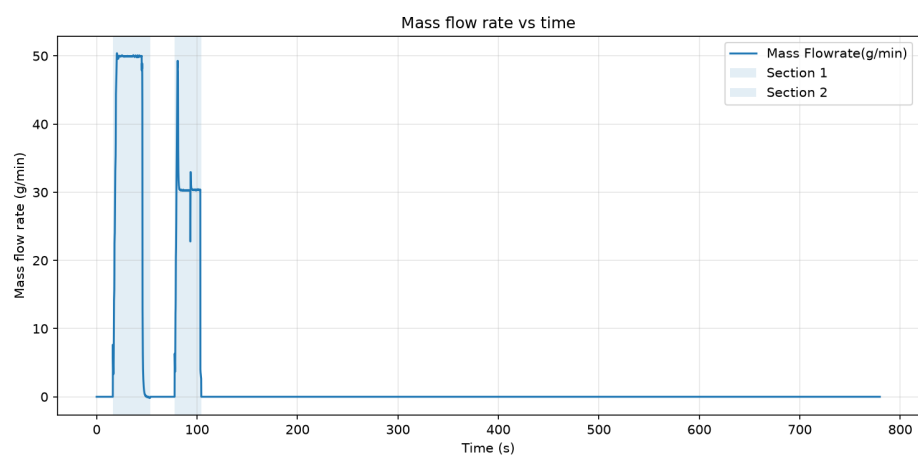


Figure 6: Mass flow rate during the Part 1b slow expansion.

2.1.2 Tank Volume Ratio

2.1.3 Heat Transfer and Net Heat Transfer

2.1.4 Path Independence of State Properties

2.2 Part 2: Initial Mass and Volume of the Left Tank

2.2.1 Processed Pressure, Temperature, and Mass Flow Results

2.2.2 Initial Mass and Tank Volume

2.2.3 Sources of Error and Limitations

2.2.4 Effect of Air Compressibility

3 Conclusion

References